Scotty: Learning Fast Typed Decisions from Synthetic Supervision

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Abstract. Many applications need a decision rather than a generated explanation. We introduce Scotty, a ModernBERT encoder with a Laya-style option-scoring head, trained on 18,092 Beam-generated and Typed Decisions examples. It reaches 83.03% accuracy on a whole-family Beam holdout and 75.45% on the official Typed test. A matched-update Typed-only control reaches 56.34% and 78.35%, respectively: mixed supervision broadens coverage across these two collections, with a specialization trade-off. Released root Laya reaches 53.51% on the matched Beam evaluation. A batch-one Spark microbenchmark takes 23.36 ms including tokenization. These single-seed results motivate compact decision models, not universal superiority or calibrated deployment.

1 Introduction

Routing requests, checking conditions and assigning severity need decisions, not explanations. A large teacher can supply examples for a small single-pass encoder, avoiding text generation and parsing. This is data-level knowledge transfer [1], not teacher-layer extraction.

We introduce Scotty, a 421-million-parameter model combining a pretrained ModernBERT encoder [2] with the typed option-scoring design of Laya [3]. Its encoder starts from the public ModernBERT checkpoint, and its decision head is freshly initialized. No trained Laya weights are loaded. Boolean decisions, categorical choices and ordinal scores share one interface and one scoring mechanism.

Our contribution is an implemented model and evaluated training recipe, not a new Transformer architecture. We combine hard Beam-generated labels with soft Typed Decisions supervision. A matched-update, Typed-only control tests this recipe against specialization: broader coverage across two collections comes with lower accuracy on the Typed test.

2 Architecture and decision interface

Inputs are a state s, instruction q and candidates C. The output is a distribution over the K candidates.

The model adopts Laya's option-marker architecture [3]. State, question and options are jointly serialized into a token sequence. Each option is associated with a `[MASK]` marker. ModernBERT supplies contextual token representations. A learned three-type embedding is added to every token before two norm-first

Transformer layers (width 1,024; 16 heads; feed-forward width 4,096; dropout 0.1). Option-marker states pass through a shared scalar scorer:

$$H = F_\theta(\text{serialize}(s, q, C)), \tag{1}$$

$$u_k = G_\phi(H + e_{\text{type}})_{m_k}, \tag{2}$$

$$z_k = W_2\ \text{GELU}(W_1\,\text{LN}(u_k) + b_1) + b_2, \tag{3}$$

$$p_k = \frac{\exp(z_k)}{\sum_{j=1}^{K} \exp(z_j)}. \tag{4}$$

Here m_k is option k's marker position, F_θ is ModernBERT, and G_ϕ is the Transformer head. Shared scoring permits variable candidate counts rather than a fixed classifier vocabulary.

Categorical decisions use $\arg\max_k p_k$; Boolean decisions expose the probability of the true option. Ordinal candidates have ordered values v_k, supporting either an argmax label or the expected score $\sum_k p_k v_k$.

The network has 421,293,827 parameters, including a 26,512,131-parameter head. An inherited 263,938-parameter auxiliary action module is unused by the decision loss. The encoder and decision path are trainable.

Serialization is `[CLS] <type> question: <instruction> [SEP]`, followed by marker–option pairs, `[SEP]`, state and `[SEP]`. Options are capped at 48 tokens each, the question/options section at 256, and the full sequence at 1,024. Boolean options are canonicalized to false/true; ordinal options follow their explicit order. Rationales, metadata and labels are excluded. Joint encoding means candidate scores cannot be independently chunked without changing the computation.

3 Training data and partitioning

3.1 Beam-generated decisions

The synthetic source contains 19,878 accepted decisions across 18 operational task families. The teacher was Reflection Beam 1.0 501B-A23B [4], accessed through Hugging Face Inference Providers. Generation used temperature 0.65, high reasoning effort and JSON responses; one initial pilot used maximum effort. Requests contained one family and option count, with up to six examples and round-robin target positions. Boolean tasks had two options, categorical tasks three to six, and ordinal tasks four to six.

Filtering used schema and answer checks, exact hashes, lexical screening (word-trigram Jaccard threshold 0.82), and ModernBERT similarity screening followed by an assistant-assisted offline content audit. Reviewed duplicate edges linked related examples. This is not exhaustive semantic or label validation: only 21 of 3,878 lower-confidence neighbor edges were sampled for review. The corpus uses API-generated data, not a locally quantized teacher.

The protected whole-family test contains 3,259 examples: operational evidence strength (935), packing readiness (1,146), and warehouse exception routing (1,178). The other 15 families supply 16,619 examples, split into 13,292

training and 3,327 validation items. Generation requests and reviewed duplicate links form components constrained to one split; allocation uses neither targets nor model performance. Validation shares families with training, whereas the test holds out three entire families. Broader conceptual overlap is still possible.

3.2 Typed Decisions

Public Typed Decisions [5] (revision `468b1461d59e`) covers customer service, invoices, security and agent traces. Cases contain multiple questions with teacher target distributions. The official 1,200 training cases (6,000 decisions) are workflow-stratified into 960 training cases and 240 validation cases; questions from a case remain together. The official test has 400 cases and 2,000 decisions.

Source	Training	Validation	Test
Beam-generated	13,292	3,327	3,259
Typed Decisions	4,800	1,200	2,000
Total decisions	18,092	4,527	5,259

Table 1: Separate source partitions. Test metrics are reported by source, not pooled into a single score. Typed Decisions is split at the case level.

The training union contains 6,019 Boolean, 6,044 categorical and 6,029 ordinal decisions. Balanced type counts do not guarantee semantic coverage. Source identities and original datasets are retained; neither test set contributes gradients, calibration fitting or checkpoint selection.

4 Learning and evaluation protocol

4.1 Learning from hard and soft targets

For a decision with target distribution t, we minimize cross-entropy:

$$\mathcal{L}(\theta, \phi) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K_i} t_{ik} \log p_{ik}. \tag{5}$$

Beam targets are one-hot; Typed targets retain supplied probability distributions. Beam token logits are unavailable: transfer is label-level. We use no ordinal-specific loss or label smoothing.

Training starts from pretrained `answerdotai/ModernBERT-large` (revision `45bb4654`) and a fresh seed-0 head. We use AdamW, encoder/head learning rates $2.5 \times 10^{-5}/10^{-4}$, weight decay 0.01, gradient clipping 1.0, and cosine decay to 10^{-6} without warm-up. Micro-batch size eight with eight-step accumulation gives effective batch 64. Full fine-tuning uses BF16 autocast and gradient checkpointing, with natural source proportions and no resampling or option shuffling. Three epochs yield 849 optimizer updates.

Selection maximizes the mean of Typed validation soft agreement and Beam validation position-balanced accuracy. Epoch three is selected. Reloaded validation scores are 91.13% Beam accuracy (92.63% balanced) and 78.33% Typed accuracy (61.62% soft agreement). No RLCD or calibration fitting is performed.

4.2 Matched-update Typed-only control

The control uses the same pretrained encoder, fresh seed-0 head, optimizer, schedule and other hyperparameters, but only the 4,800 Typed training decisions from 960 cases. Cyclic shuffled passes match Scotty's 849 updates and 54,276 decision exposures. Thus the control repeats Typed examples more often; unique data and source exposure differ. Equal updates and examples do not imply equal tokens or FLOPs. Validation at updates 283, 566 and 849 uses the same joint selection rule; update 849 is selected. Beam validation therefore informs control selection, although no Beam training item contributes gradients. This is a retrospective controlled ablation: both tests had already been examined in the project.

4.3 What the evaluations measure

Released root Laya is evaluated on the same 3,259 family-held-out examples with matched conversion, precision, batching and scoring, but not matched training budget. Its original training-data overlap is unknown. This test was protected during Scotty training, but used in earlier project evaluations; it is not unseen by the entire research process.

Typed Decisions tests unseen cases in trained workflows, not zero-shot domain transfer. Normalized exact overlap with training and validation is zero. All 2,000 test decisions are supported without truncation or collapsed markers. The checkpoint remains frozen.

Hard accuracy compares model and target argmax. Position-balanced accuracy macro-averages observed (type, option count, converted gold position) cells. Soft agreement averages the teacher probability assigned to the predicted option. For ordinal questions, we also report mean absolute error in scale positions. Uncertainty on Typed Decisions accuracy is estimated by resampling whole cases, preserving within-case dependence. Confidence intervals describe test-sample variability for this trained model, not variability across training seeds.

5 Experimental results

5.1 Accuracy and comparison with Laya

On the synthetic holdout, Scotty reaches 83.03% accuracy, versus 53.51% for released root Laya (Table 2). Position-balanced accuracy is 76.39% versus 48.56%. The lower balanced scores caution against relying on raw accuracy alone. The raw difference is 29.52 percentage points; the model's scenario-clustered 95% interval is 81.29–84.75% (543 request clusters, 10,000 bootstrap draws). This comparison measures checkpoint performance, not an isolated data effect.

Evaluation	Model	Accuracy	Balanced
Beam holdout (measured)	Root Laya	53.51	48.56
	Typed-only control	56.34	51.64
	Scotty	**83.03**	**76.39**
Typed test (measured)	Typed-only control	**78.35**	–
	Scotty	75.45	–
	Majority baseline	48.15	–
Typed test (published [6])	Root Laya	36.20	–
	Laya Typed Decisions	76.60	–

Table 2: Accuracy in percent. Balanced means the position-balanced metric for the Beam holdout. Published Typed results are context, not matched reruns; their adapters and training procedures differ.

Typed hard accuracy is 75.45% (case-clustered 95% interval: 73.30–77.60%). A training-derived majority rule achieves 48.15%; expected uniform chance is 31.75%. Published specialized Laya accuracy is 76.60%, but this cross-source comparison establishes neither statistical equivalence nor superiority.

5.2 Coverage versus specialization

Against the matched-update control, Scotty gains 26.70 percentage points on Beam raw accuracy (paired request-cluster 95% interval: 24.96–28.42). Conversely, the control gains 2.90 points on Typed accuracy (paired case-cluster interval: 1.05–4.80) and reaches 61.11% soft agreement versus 60.20%. Differences use unrounded scores. These results support a coverage–specialization tradeoff, not a claim that synthetic supplementation improves Typed performance.

On Beam Boolean, categorical and ordinal tasks, Scotty scores 90.75%, 91.77% and 62.57%, versus the control's 54.80%, 74.96% and 34.76%. Its Typed scores are 83.17%, 72.67% and 71.75%, respectively. Ordinal decisions remain challenging: Typed ordinal MAE is 0.305 scale positions, with 97.88% within one level. The ablation supports the mixed recipe on the Beam holdout, but does not separate added task coverage from greater unique-example diversity.

5.3 Probabilities and inference cost

Typed soft agreement is 60.20%; its attainable teacher-argmax bound is 65.89%, not a hard-accuracy ceiling. Raw 15-bin expected calibration error is 0.1093. No calibration was fitted; probabilities and deployment thresholds require a separate calibration protocol [7].

On Spark's NVIDIA GB10, BF16 batch-one inference for one 350-token, four-option validation item takes 22.21 ms median (23.09 ms p95; 20 runs), or 23.36 ms including tokenization/collation (23.64 ms p95; ten runs). Five warm-ups precede synchronized GPU timing. Network overhead is excluded; concurrency was unlogged. This is not a throughput or matched LLM comparison.

6 Discussion and conclusion

A pretrained encoder and shared typed head support fast decisions across two collections. The control sharpens the conclusion: mixed supervision improves Beam accuracy, while Typed-only training better specializes on Typed Decisions. Neither dominates both tests.

The study uses one seed and retrospectively inspected tests. Clustered intervals capture test-sample uncertainty, not training variability. Matching updates and exposures is not a compute-matched comparison, and joint validation includes Beam information for both models. Both sources are synthetic; unrelated real-world generalization and pretraining contamination remain unresolved. Teacher labels are not guaranteed truth. Calibration, domain review and validated escalation are needed before deployment.

Additional seeds, token/FLOP-aware controls and independently constructed tests would strengthen the evidence. New data should address development-set coverage gaps, not consumed test errors. Within these limits, Scotty offers a practical route from expensive supervision to compact local decision interfaces.

References

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